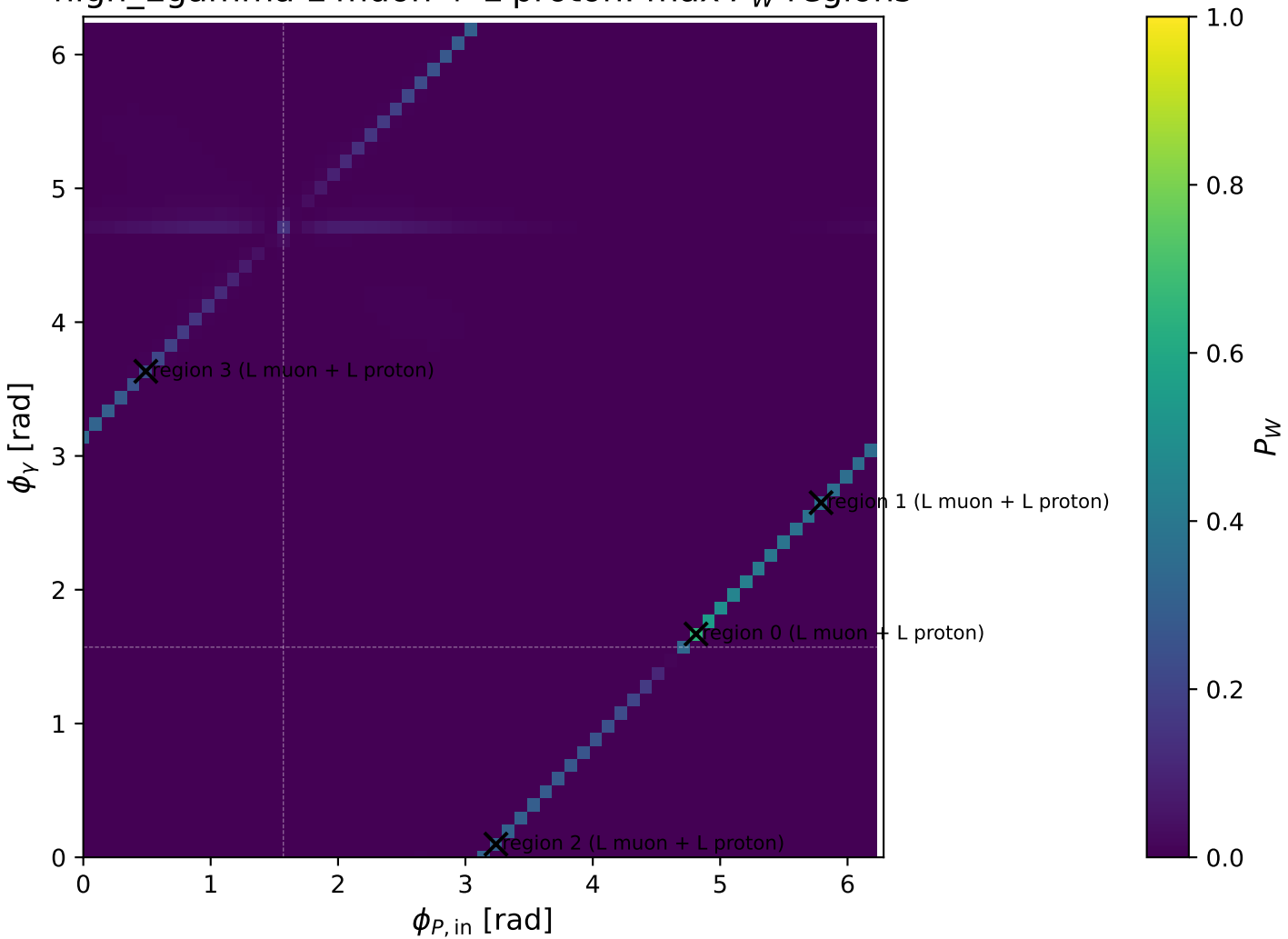
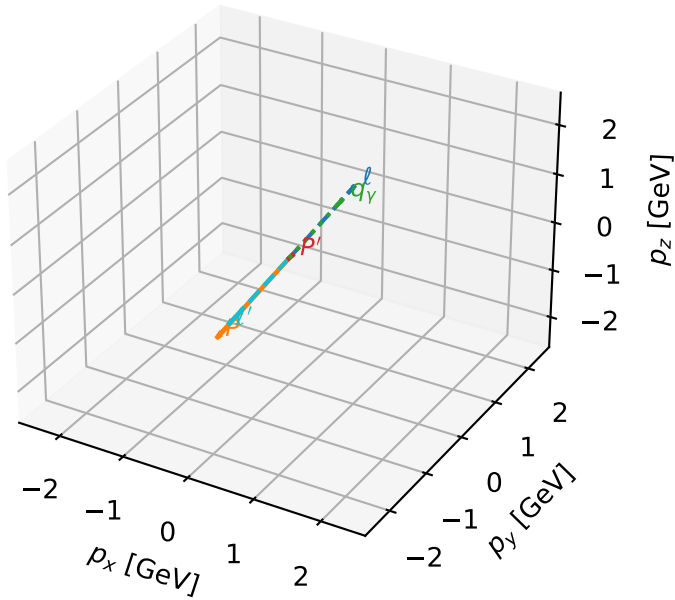


high_Egamma L muon + L proton: max P_W regions



L muon + L proton characteristic max P_W configuration

3D momenta

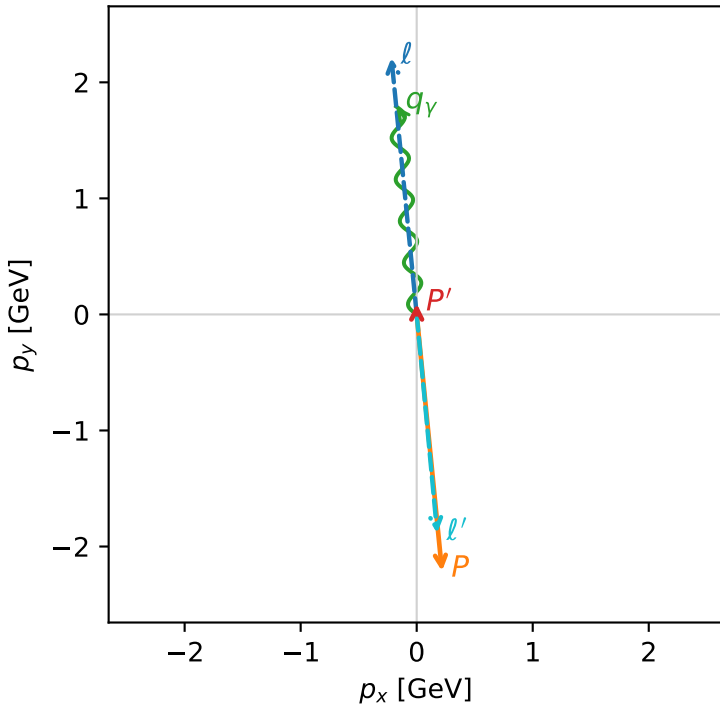


w_purity_high_egamma_ll_region_0 P_W=0.651136
 region: high_Egamma_LL_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

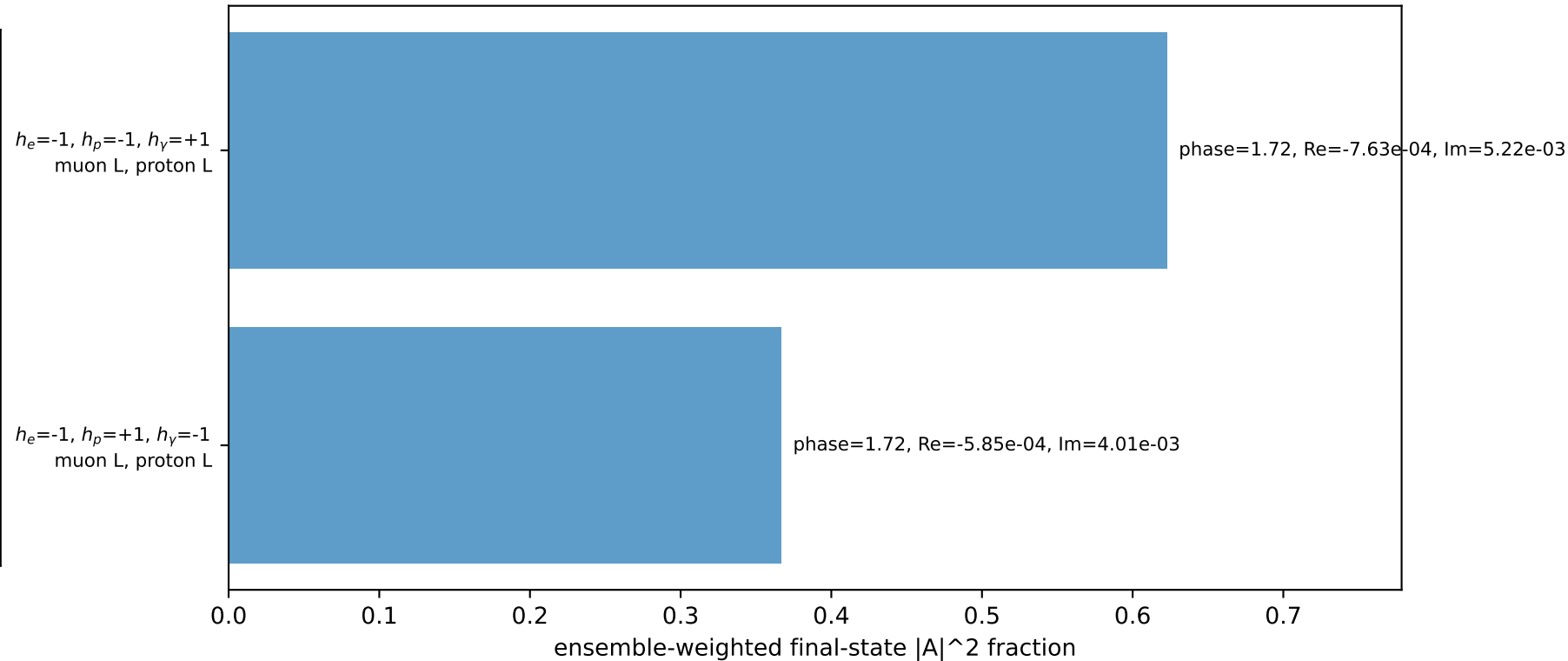
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.0933634$
 $\theta_{in}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$
 $E_\gamma=1.8$, $\phi_\gamma=1.66897$
 $Q^2=16.833$, $x_B=0.84622$, $t=-3.20239$, $W^2=3.93884$, $y=0.964469$
 $\theta(l', \gamma)=179.723$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 l' $E= 1.8959$ $p_x= 0.1764$ $p_y= -1.8847$ $p_z= 0.0000$
 P' $E= 0.9426$ $p_x= 0.0000$ $p_y= 0.0934$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.1764$ $p_y= 1.7913$ $p_z= 0.0000$

Transverse plane

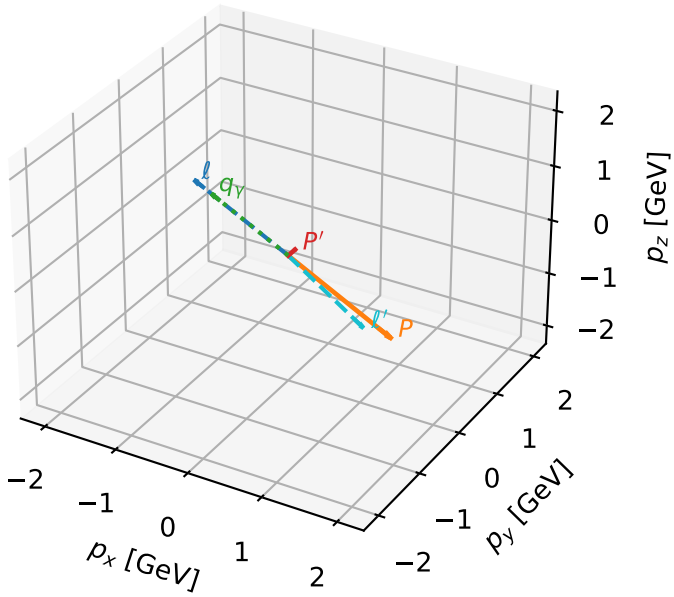


Leading initial-ensemble/final-state components (N=2, retained=99.0%)



L muon + L proton characteristic max P_W configuration

3D momenta



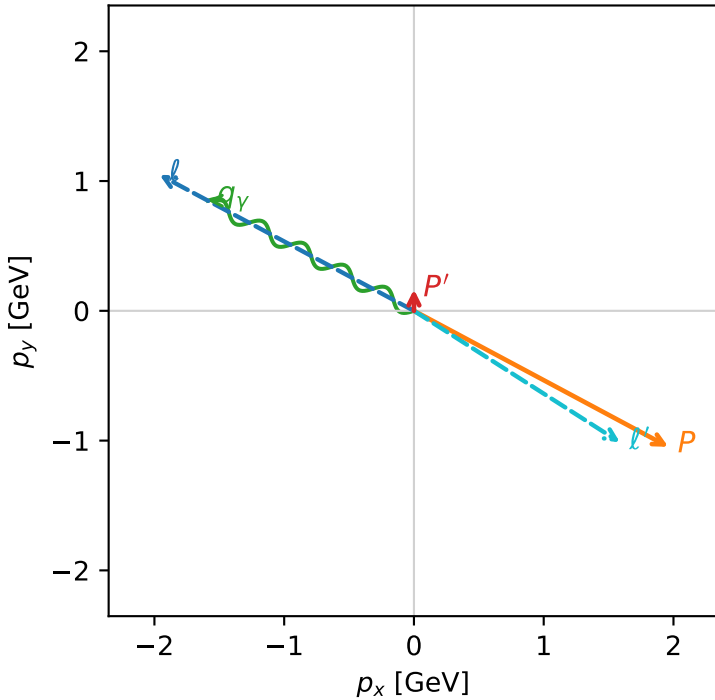
w_purity_high_egamma_ll_region_1 P_W=0.36582
 region: high_Egamma_LL_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.16464$
 $\theta_{in}=1.57079$, $\phi_l=2.65072$, $\phi_p=5.79231$
 $E_\gamma=1.8$, $\phi_\gamma=2.65072$
 $Q^2=16.7218$, $x_B=0.841525$, $t=-3.18118$, $W^2=4.02887$, $y=0.963441$
 $\theta(l', \gamma)=175.578$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

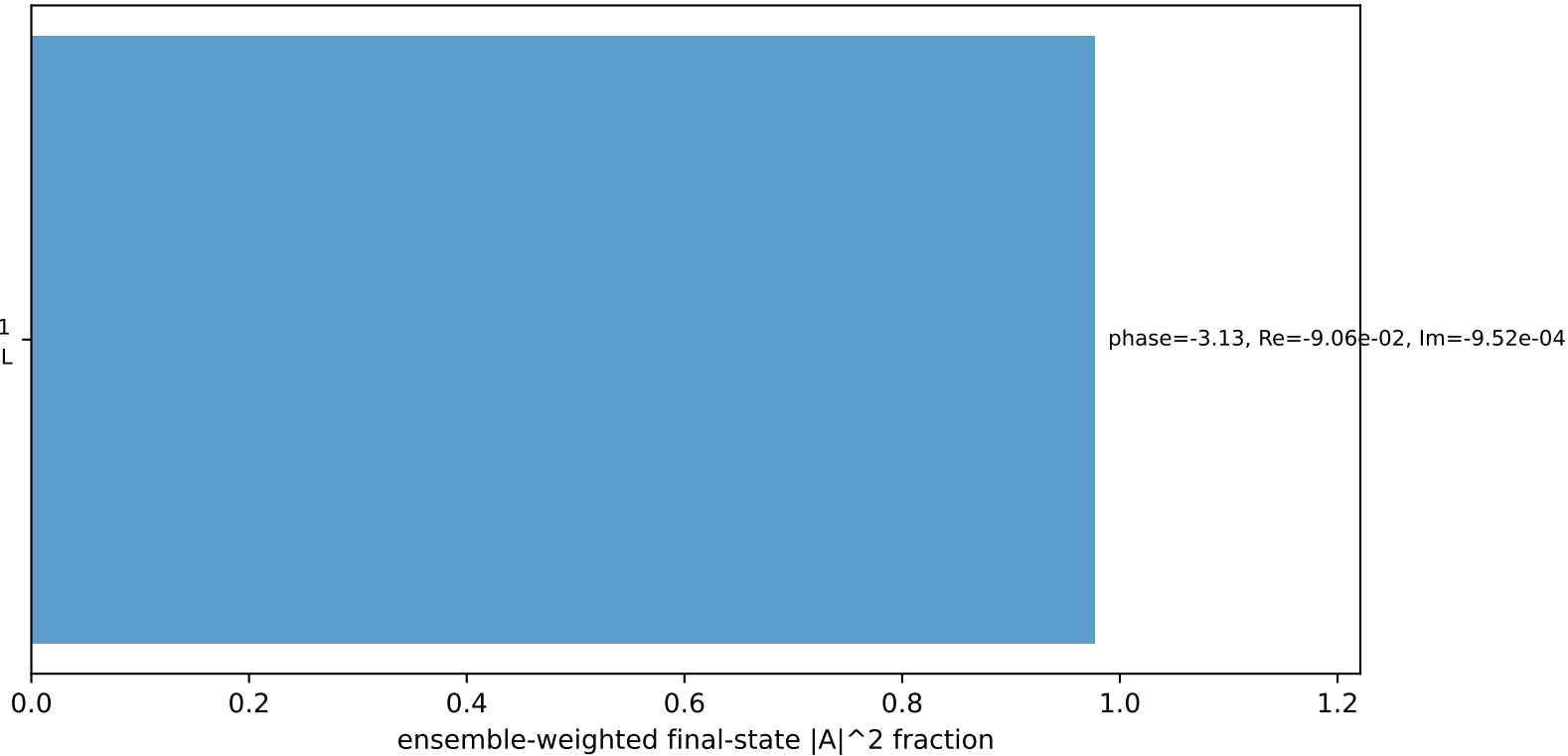
l $E= 2.2256$ $p_x= -1.9606$ $p_y= 1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= -1.0480$ $p_z= 0.0000$
 l' $E= 1.8862$ $p_x= 1.5875$ $p_y= -1.0132$ $p_z= 0.0000$
 P' $E= 0.9523$ $p_x= 0.0000$ $p_y= 0.1646$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.5875$ $p_y= 0.8485$ $p_z= 0.0000$

Transverse plane



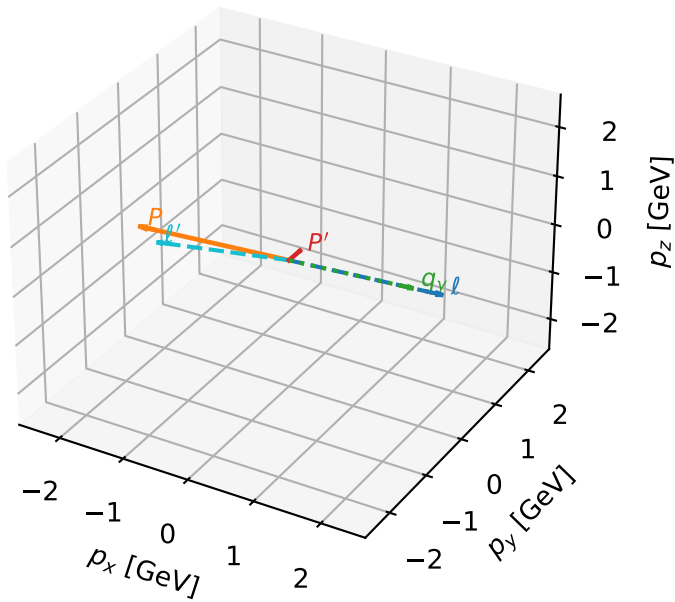
$h_e=-1$, $h_p=-1$, $h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=1, retained=97.7%)



L muon + L proton characteristic max P_W configuration

3D momenta

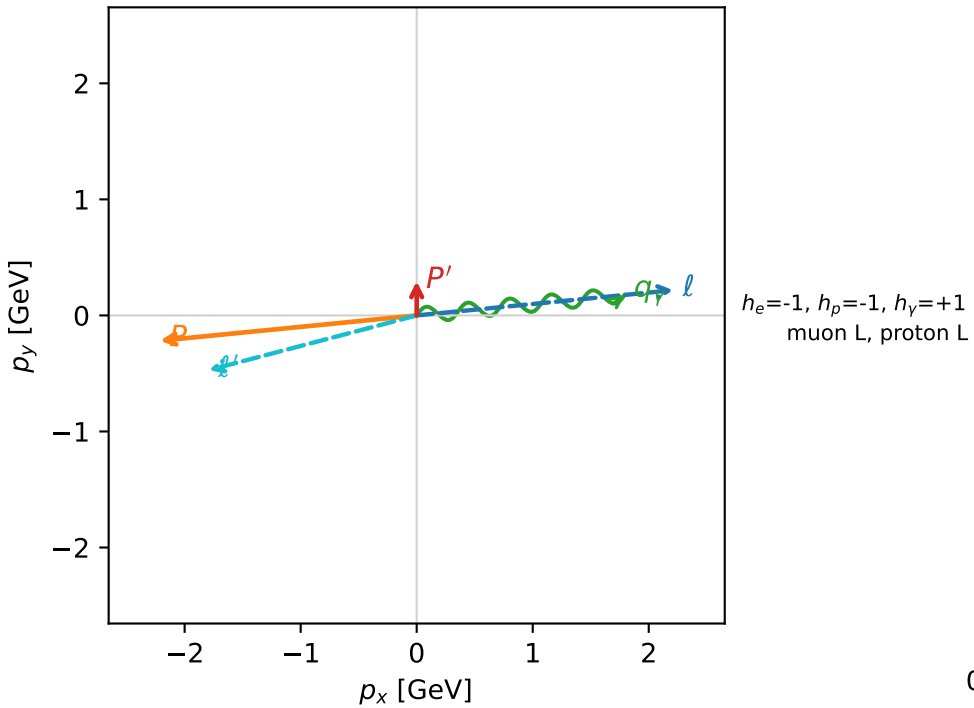


w_purity_high_egamma_ll_region_2 $P_W=0.311226$
 region: high_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

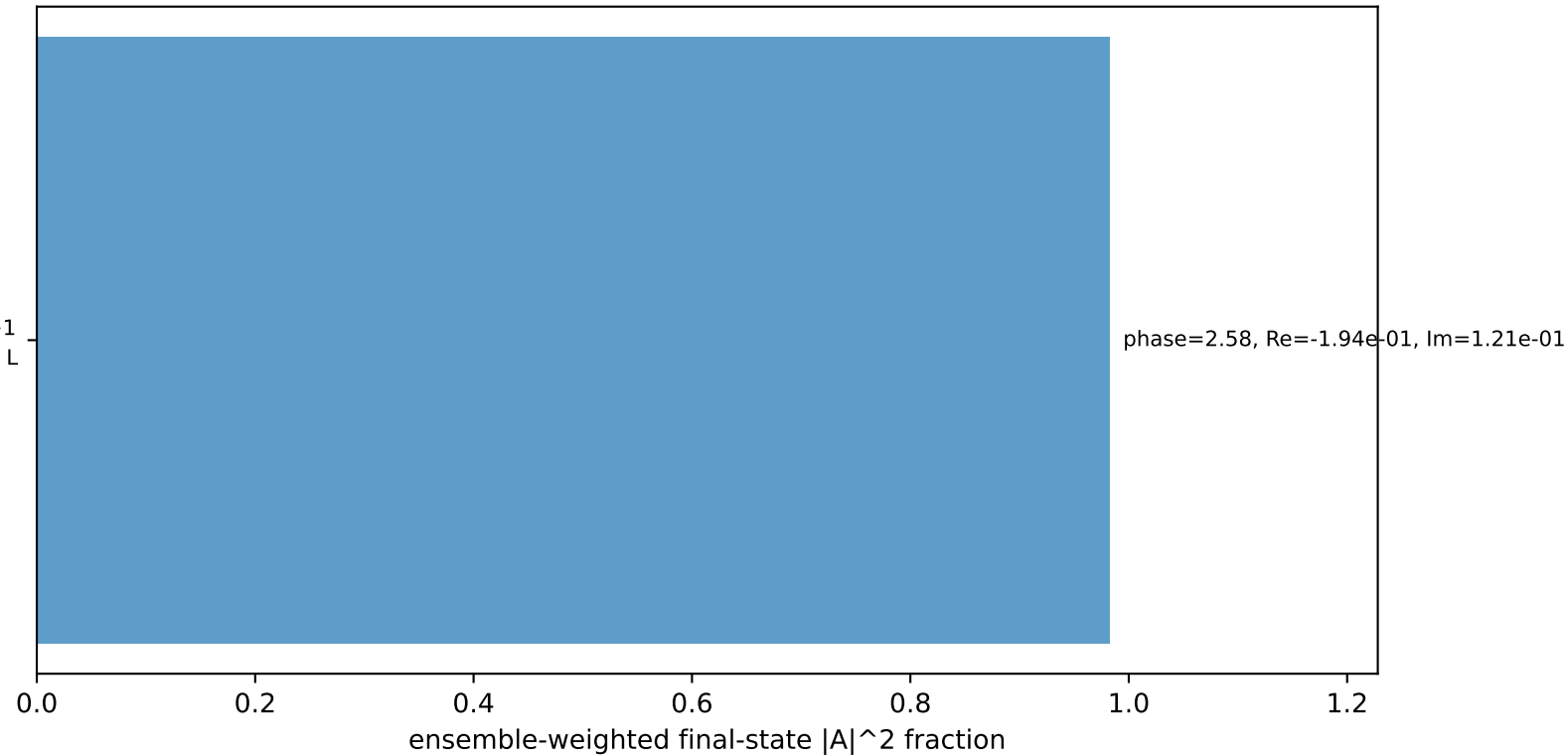
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.29479$
 $\theta_{in}=1.57079$, $\phi_l=0.0981748$, $\phi_p=3.23977$
 $E_\gamma=1.8$, $\phi_\gamma=0.0981748$
 $Q^2=16.3677$, $x_B=0.826513$, $t=-3.11366$, $W^2=4.31546$, $y=0.960167$
 $\theta(l', \gamma)=170.887$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 2.2124$ $p_y= 0.2179$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -2.2124$ $p_y= -0.2179$ $p_z= 0.0000$
 l' $E= 1.8553$ $p_x= -1.7913$ $p_y= -0.4712$ $p_z= 0.0000$
 P' $E= 0.9832$ $p_x= 0.0000$ $p_y= 0.2948$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 1.7913$ $p_y= 0.1764$ $p_z= 0.0000$

Transverse plane

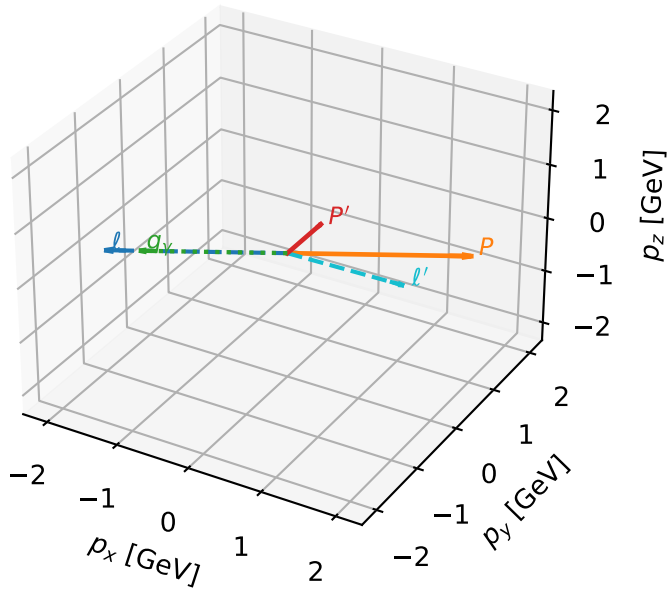


Leading initial-ensemble/final-state components (N=1, retained=98.2%)



L muon + L proton characteristic max P_W configuration

3D momenta



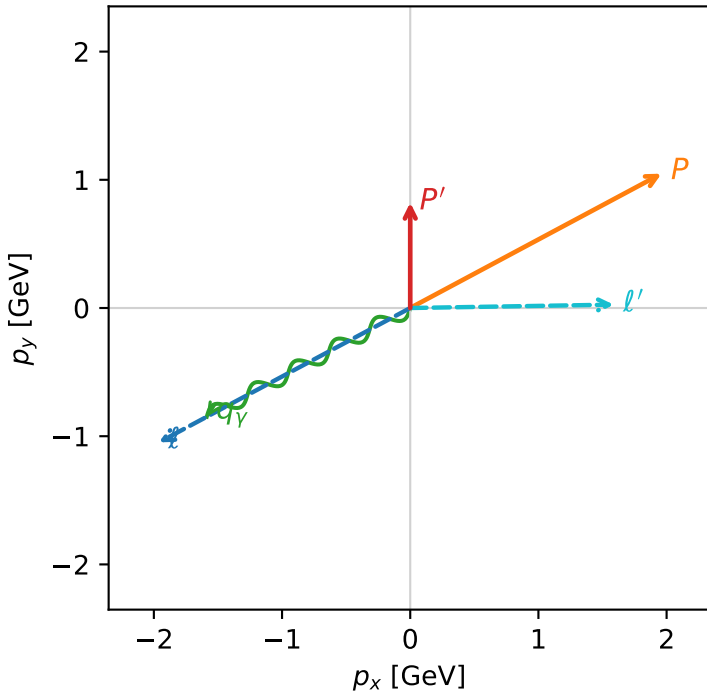
w_purity_high_egamma_ll_region_3 P_W=0.233969
 region: high_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.822183$
 $\theta_{in}=1.57079$, $\phi_l=3.63247$, $\phi_p=0.490874$
 $E_V=1.8$, $\phi_V=3.63247$
 $Q^2=13.3404$, $x_B=0.693871$, $t=-2.53642$, $W^2=6.7655$, $y=0.932179$
 $\theta(l', \gamma)=152.825$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= -1.9606$ $p_y= -1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= 1.0480$ $p_z= 0.0000$
 l' $E= 1.5912$ $p_x= 1.5875$ $p_y= 0.0263$ $p_z= 0.0000$
 P' $E= 1.2473$ $p_x= 0.0000$ $p_y= 0.8222$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.5875$ $p_y= -0.8485$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 muon L, proton L

$h_e=-1, h_p=+1, h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=2, retained=99.6%)

